Course:	COMP 2131	**Written Assignment-2**
Weight:	**100 marks =** 8% of final grade	**Memory Hierarchy/Linking/Assembly**

Instructions

Complete the questions given below for Module 4, Module 5 and introduction to Module 6. Enter your answers directly into this document using red text. Submit it to the Open Learning Faculty Member after completing all the questions.

Module-4 [40 marks]

Q1. Calculate the access time for a disk having the size of 512 byte/sector and the seek time of 12ms. The rotational rate is 5700 RPM. The data transfer rate is 3MB/sec, and the the controller overhead is 1ms, you may even assume that there is no service time for the access. [20]

Q2. The three functions as given below perform the same operation with varying degrees of spatial locality. Rank-order the functions with respect to the spatial locality enjoyed by each. [10]

(a) An array of structs

```
#define N 1000

typedef struct {
    int vel[3];
    int acc[3];
} point;

point p[N];
```

Explain how you arrived at your ranking.

```
void clear3(point *p, int n)
{
    int i, j;

    for (j = 0; j < 3; j++) {
        for (i = 0; i < n; i++)
            p[i].vel[j] = 0;
        for (i = 0; i < n; i++)
            p[i].acc[j] = 0;
    }
}
```

(b) The clear1 function

```
void clear1(point *p, int n)
{
    int i, j;

    for (i = 0; i < n; i++) {
        for (j = 0; j < 3; j++
            p[i].vel[j] = 0;
        for (j = 0; j < 3; j++
            p[i].acc[j] = 0;
    }
}
```

```
void clear2(point *p, int n)
{
    int i, j;

    for (i = 0; i < n; i++) {
        for (j = 0; j < 3; j++)
            p[i].vel[j] = 0;
            p[i].acc[j] = 0;
        }
    }
}
```

Q3. Even though cache speeds up the process, larger caches tend to increase the hit time. Explain why? [10]

Module-5 [30 marks]

Q1. Explain the concept of symbol table. How does the linker use the symbol table to create executable object files? [10]

Q2. Explain the term relocation. Explain the structure of Typical ELF executable object file. [10]

Q3. What would be the output on the terminal when a user tries to compile and execute the following files. Write the steps to compile and run as well. [10]

```
#include <stdio.h>
int buf[]={1,3};
int main()
{
  swap();
printf("x= %d , y=%d\n", buf[0], buf[1]);
  return 0;
}
```

main.c

```
#include <stdio.h>
extern int buf[];

int *bufp0 = &buf[0];
static int *bufp1;
void swap()
{
  int temp;
  bufp1 = &buf[1];
  temp = *bufp0;
  *bufp0 = *bufp1;
  *bufp1 = temp;
}
```

swap.c

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Module-6 [30 marks]

Q1. In 'C' we may create register variables to speed up the process. How many register variables should we define and how ill they help in speeding up the process. Explain the answer using the knowledge of Assembly language. [10]

Q2. Assume the following values are stored at the indicated memory addresses and registers [10]

Address	Value
0x100	0xaaa
0x104	0x123
0x108	0x12
0x10c	0x10

Register	Value
%eax	0x100
%ecx	0x1
%edx	0x3

Fill up the following table:

%eax 0x104 $0x108 (%eax) 4(%eax) 9(%eax,%edx) 260(%ecx,%edx)	

0xFC(,%ecx,4)	
(%eax,%edx,4)	

Q3. For the structure declaration:

```
struct {
char *a;
short b;
double c;
char d;
float e;
char f;
long long g;
void *h;
} foo;
```

Suppose the structure was compiled on a 32-bit Windows or a 32-bit UNIX machine, where each primitive data type of K bytes must have an offset that is a multiple of K.

A. What are the byte offsets of all the fields in the structure?

B. What is the total size of the structure?

C. Rearrange the fields of the structure to minimize wasted space, and then show the byte offsets and total size for the rearranged structure. [10 marks]